

Discovery report for OpenADMET + ExpansionRx Blind Challenge prompt 2

Research Objective

Develop an excellent ML model for predicting the following properties:

LogD: Measures compounds lipophilicity at a specific pH. Drugs typically fall into a well defined LogD range that balances aqueous solubility with membrane permeability, making understanding changes in LogD across a chemical series important to medicinal chemists. Additionally, assessing LogD of candidate molecules can suggest whether candidate molecules are "efficient" for their lipophilicity (lipophilicity generally has a linear relationship with affinity). Kinetic Solubility (KSOL): Measures how much a compound can be dissolved under non-equilibrium conditions. Helps screen compounds that will fail due to poor absorption or low bioavailability. Human Liver Microsomal (HLM) stability: Supports understanding of a compound's susceptibility to liver metabolism and can be used to predict in-vivo clearance of a candidate molecule. Measured using human liver microsomes and reported as intrinsic clearance for the compound Clint (mL/min/kg). Mouse Liver Microsomal (MLM) stability: Also used to predict in-vivo clearance of a candidate. The study of both MLM and RLM can provide a more comprehensive understanding of a compound's metabolic profile and how a compound might behave in multi-species preclinical development. Caco-2 Papp A>B: Measures the rate of flux of a compound across polarized Caco-2 cell monolayers from the apical (intestinal lumen-facing side) to basolateral (blood-facing side). This effectively mimics the absorption of a drug across the intestinal wall. Caco-2 Efflux Ratio: Measures the rate of flux of a compound across polarized Caco-2 cell monolayers. The Efflux Ratio is determined by a ratio of the apparent permeability coefficient (Papp) in both directions. Ratios of ~1 indicate that a compound primarily traverses the cell membrane via passive (diffusional) transport. Ratios > 2 generally indicate active transport of the compound across the cellular membrane by membrane bound transporters (e.g. efflux by p-glycoprotein). Mouse Plasma Protein Binding (MPPB): Determines the concentration of free drug in plasma (as % Unbound). Drugs that are not bound to plasma can bind to target proteins and yield the desired therapeutic effect, making this parameter crucial to understanding drug distribution. Mouse Brain Protein Binding (MBPB): This measures the fraction of a drug not bound to proteins within brain tissue. The unbound fraction of a drug in the brain is considered pharmacologically active and able to interact with central nervous system (CNS) targets. MBPB helps assess CNS drug exposure and potential efficacy or side effects for neuroactive compounds. Reported as % Unbound. Mouse Gastrocnemius Muscle Binding (MGMB):. This reflects the amount of drug free to act within skeletal muscle tissue, which is important for drugs targeting peripheral or muscular conditions (very important for the DM1 indication). Reported as % Unbound

While training the model, pay close attention to timing information. You are running in a virtual machine with ~32GB of RAM and ~8 CPUs and 1200 seconds per cell timeouts.

You must have predictions in a csv (same format as training) for the test compounds. As soon as possible (after a few iterations), start creating predictions files from the best current models.

On the leaderboard, the best scores are 0.81 spearman/0.53 MA-RAE. You should do a literature search (after dataset exploration, because chemistry tasks require some specific featurization, model architectures, data splitting, and training strategies). Although be careful about compute time!

You have been provided as datasets information about the ADMET properties and the context in which the data were collected. Use that to guide modeling decisions. You have also been provided information about other methods that have performed well and poorly on this dataset, including the report from a high-quality Kosmos run. Use that to guide the modeling decisions. You are also provided exploratory data analysis code that you can use to guide your own exploratory analyses if that is useful. If there are existing chemical informatics or machine learning models that are available to install as packages or download from accessible repositories, use them. However, be careful about

your computational resource constraints, ability to access internet resources, and ability to install software in a sandboxed environment.

Dataset Description

The dataset contains the following files:

advancing-predictive-ADMET-modeling.pdf - a PDF with context about the dataset being modeled.
announcement-2-ExpansionRx-OpenADMET-blind-challenge.pdf - a PDF with context about the dataset being modeled.

currentleaderboard__202512__05.csv - the leaderboard with the rank (1 is best), performance metrics, and optional model details of the current models. When model details are provided, they may be a URL to a research manuscript or a GitHub repository.

expansion_data__testblinded.csv - the compounds for which predictions must be made as SMILES strings.

expansion_data__train.csv - the training dataset with SMILES strings and the ADMET properties.

expansion_data__train_raw.csv - an alternative format of the training dataset in raw form.

expansion_tutorial.ipynb - a Jupyter notebook showing exploratory data analysis of the dataset.

Expansion-CDD-webinar.pdf - a PDF with context about the dataset being modeled.

kosmosaw1-run-20251114-1120-repl3.pdf - the report generated from a previous high quality Kosmos analysis of this dataset. Learn from this analysis and improve on it.

MacDermottOpeskin2025.pdf - a manuscript describing a similar competition to predict ADMET properties. Learn about the dataset and methodology in that competition and use it to guide modeling decisions.

openadmet_expansion__exploration.py - a Python script showing exploratory data analysis of the dataset.

OpenADMET-ExpansionRx-Blind-Challenge-about.pdf - a PDF with context about the dataset being modeled.

OpenADMET-ExpansionRx-Blind-Challenge-FAQ.pdf - a PDF with context about the dataset being modeled.

openadmet-team-photo.jpg - an image of the competition organizers.

Summary of Discoveries

Discovery 1: Structure-aware validation and dataset topology govern generalization in the ADMET Expansion benchmark

The ADMET Expansion benchmark exhibits high-quality but sparsely observed multi-assay measurements and a topology where every molecule has a unique scaffold, making conventional scaffold splits effectively random. Structure-aware validation via Butina clustering yields lower, more realistic generalization estimates and reveals that performance depends on assay and molecular complexity, with multi-task neural networks and fingerprint features providing the strongest overall results.

Discovery 2: Hybrid ensembling with task isolation mitigates negative transfer and optimizes ADMET prediction

Across nine ADMET endpoints with structured sparsity, a simple yet principled hybrid ensemble that isolates highdata tasks from a shared multitask neural network and blends them with optimized singletask gradient boosting achieves stateoftheart crossvalidated accuracy. The approach mitigates negative transfer in multitask learning and shows that taskspecific weighted averaging outperforms more complex ensembling strategies like stacking or fully dynamic weighting.

Discovery 3: Numerically stable multi-task training and universal log-transformations yield physically consistent, higher-accuracy models

This work identifies and resolves a subtle, yet consequential, source of training instability in multitask

neural networks for ADMET prediction and shows that enforcing physical constraints in target space via a universal log1p transformation is superior to posthoc output clipping or BoxCox power transforms. Together, numerically stable masking and universal log1p produce physically consistent models with substantially higher rank accuracy and lower magnitudesensitive error across nine assays.

Discovery 4: Error deconvolution identifies chemotype-specific blind spots, but RDKit-level targeted augmentation rarely improves performance

Error deconvolution of a strong ADMET ensemble revealed chemotype-specific blind spotsespecially for polar, amide and aminerich moleculesand endpointspecific dependencies on lipophilicity and polarity. Despite this clarity, most RDKitlevel targeted augmentations failed to improve performance; the only reproducible gain came from a compact set of 20 physicochemical descriptors added to Morgan fingerprints in a multitask neural network, yielding modest but significant improvements, while broader or handpicked additions were largely redundant or noisy.

Structure-aware validation and dataset topology govern generalization in the ADMET Expansion benchmark

Summary

The ADMET Expansion benchmark exhibits high-quality but sparsely observed multi-assay measurements and a topology where every molecule has a unique scaffold, making conventional scaffold splits effectively random. Structure-aware validation via Butina clustering yields lower, more realistic generalization estimates and reveals that performance depends on assay and molecular complexity, with multi-task neural networks and fingerprint features providing the strongest overall results.

Background

Accurate prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) endpoints is central to medicinal chemistry because properties such as lipophilicity (LogD), solubility, microsomal clearance, epithelial permeability and efflux, and tissue- and plasma-protein binding jointly determine exposure and efficacy. These endpoints are typically measured with uneven coverage across compounds and are intercorrelated to varying degrees, which motivates multi-task learning and careful validation strategies. In this context, rigorously assessing generalization across chemical spacespecially to novel scaffolds or structural clustersis as important as maximizing cross-validation scores, because overoptimistic splits can obscure failure modes in prospective use.

Results & Discussion

The Expansion dataset comprises 5,326 training molecules and 2,282 test molecules spanning nine assays (LogD, KSOL, HLM CLint, MLM CLint, Caco-2 Papp A>B, Caco-2 Efflux ratio, and mouse plasma, brain, and muscle protein binding) with pronounced but structured sparsity: core properties such as LogD and KSOL have >94% coverage, secondary assays reach 4071%, while specialized protein binding tasks range from 424% coverage; censoring is minimal (<3% across assays), only 124 molecules have all nine endpoints, and inter-assay correlations are strong where expected (e.g., MBPBMGMB $r=0.904$; LogDMPPB $r=$

0.686; HLMMLM $r=0.561$) [r0]. Train and test share no molecular overlap and differ systematically in complexity (mean SMILES length 48.0 vs 57.8), implying an extrapolation challenge; critically, every molecule has a unique Murcko scaffold, so scaffold splits collapse to random splits, and models must cope with missing labels to exploit the dataset fully [r0, r8].

Feature comparisons show that circular fingerprints capture structureproperty relationships more effectively than low-dimensional physicochemical descriptors and are more robust to changes in molecular complexity. Across nine assays, Morgan fingerprints (radius=2, 2048 bits) outperformed a 26-descriptor panel in 8/9 tasks (e.g., LogD RMSE 0.474 vs 0.578; KSOL 74.63 vs 84.05; HLM CLint 101.22 vs 109.64), with only a marginal edge for descriptors on Caco-2 Papp A>B (7.68 vs 7.76) [r4]. The errorcomplexity relationship was heterogeneous: prediction error increased with complexity in HLM ($\rho=0.048$, $p=3.6 \times 10^{-3}$) and Caco-2 Efflux ($\rho=0.225$, $p<10^{-3}$), but decreased in MLM ($\rho=-0.232$, $p<10^{-3}$), Caco-2 Papp A>B ($\rho=-0.211$, $p<10^{-3}$), MBPB ($\rho=-0.163$, $p<10^{-3}$), and MGMB ($\rho=-0.433$, $p<10^{-3}$), underscoring that generalization difficulty is property-dependent rather than uniformly harder for complex molecules [r4]. Stratifying by BertzCT quartiles confirmed that fingerprints degraded less than descriptors in 7/9 assays, indicating greater robustness to complexity-induced distribution shift [r4].

Leveraging cross-assay structure with a single multi-task neural network (shared 2048256256128 backbone and nine task-specific heads; masked MSE and per-task z-scoring) substantially improved accuracy over a LightGBM baseline when evaluated under the datasets effective random (unique-scaffold) 5-fold split. The MTL model achieved an average Spearman correlation of 0.779 (MA-RAE 0.5337) versus 0.6589 for LightGBM, with gains in all nine endpoints including large improvements in sparse tasks (MGMB +0.245 to 0.815;

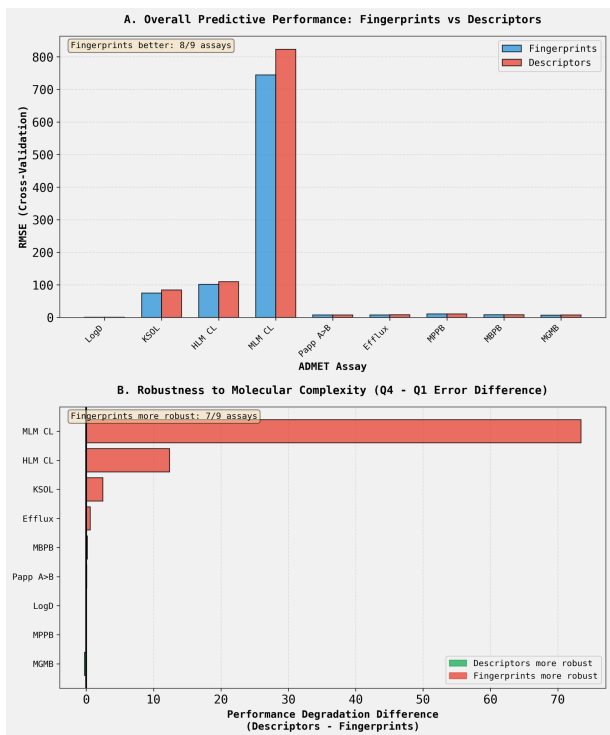


Figure 1: Fingerprint features yield higher predictive accuracy and greater robustness to molecular complexity than physicochemical descriptors. (A) Comparison of cross-validation Root Mean Square Error (RMSE) for models using either fingerprint or descriptor features across nine ADMET assays, where lower error indicates better performance. (B) The difference in performance degradation between the two feature types, calculated as the change in error between the highest (Q4) and lowest (Q1) molecular complexity quartiles. Positive values indicate that fingerprints are more robust to increasing molecular complexity. Overall, fingerprints better capture the structure-property relationships and are less sensitive to the extrapolation challenge posed by the benchmark dataset. (Source: [r4])

MBPB +0.184 to 0.844; HLM CLint +0.148 to 0.688) and consistent benefits in high-coverage assays (LogD +0.029 to 0.919; KSOL +0.034 to 0.744); all p-values were $<10^{-3}$ [r8]. These results validate masked multi-task learning as an effective strategy for structured sparsity, but the absence of shared scaffolds across folds makes these estimates optimistic with respect to novel-chemotype generalization [r8].

A structure-aware evaluation using Butina clustering (Tanimoto similarity 0.6; 948 clusters with sizes 1264) addresses this limitation and produces systematically lower, more realistic estimates while revealing assay-specific sensitivity to chemical topology. Assigning whole clusters to folds caused inevitable fold imbalance

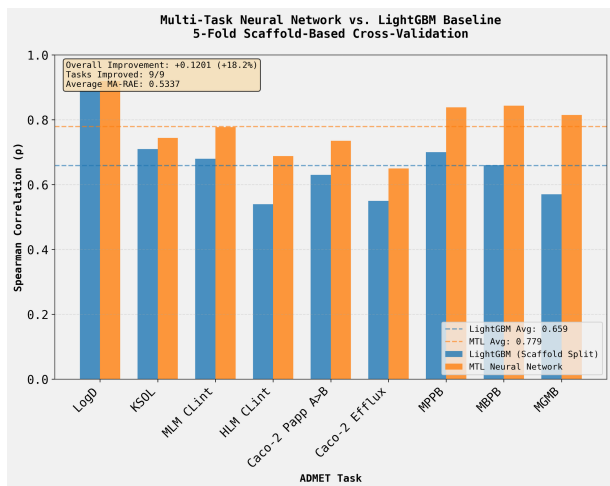


Figure 2: A multi-task neural network model consistently outperforms a LightGBM baseline across nine ADMET properties under 5-fold scaffold-based cross-validation. The bar plot compares the Spearman correlation (ρ) for the multi-task neural network and the LightGBM model on each task, with dashed lines indicating the average performance across all tasks. The consistent improvement, with an average correlation of 0.779 for the neural network versus 0.659 for LightGBM, highlights the advantage of a multi-task architecture for this sparsely labeled dataset. (Source: [r8])

and reduced performance for both model classes: relative to the unique-scaffold (random) results, mean Spearman correlations dropped by 0.0268 for LightGBM and 0.0608 for the MTL network; HLM CLint showed the largest declines (LightGBM -0.0694; MTL -0.0895), whereas MPPB was nearly unchanged (-0.0005) [r11]. Despite the larger drop, the MTL model remained superior in 6/7 evaluated tasks under cluster-based cross-validation, with absolute Spearman ranges of 0.5430.872 for MTL versus 0.4710.873 for LightGBM, reinforcing that topology-aware validation tightens headline metrics yet preserves the relative advantage of representation learning for sparse endpoints [r11]. Together with the observed complexity shift and assay-specific error/complexity trends, these results demonstrate that both dataset topology and validation design govern apparent generalization on this benchmark, motivating the combined use of fingerprint features, masked multi-task learning, and cluster-based validation to obtain both strong performance and credible estimates of prospective utility [r0, r4, r8, r11].

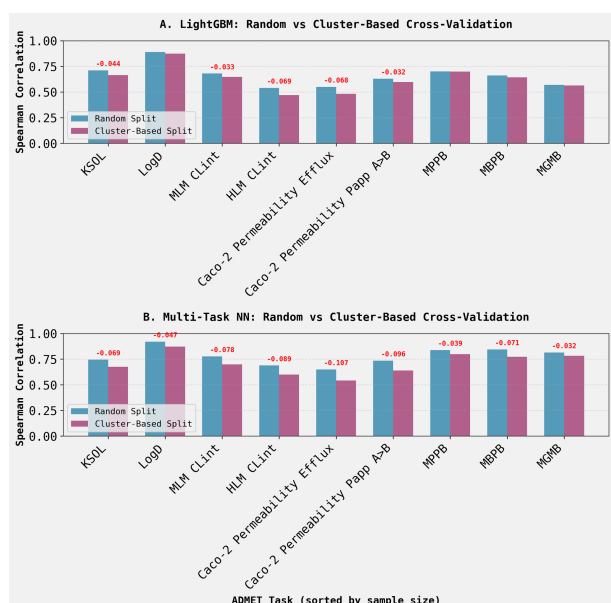


Figure 3: Structure-aware, cluster-based cross-validation consistently yields lower performance estimates than standard random splits. The plots compare the mean Spearman correlation from 5-fold cross-validation for (A) a LightGBM model and (B) a Multi-Task Neural Network across nine ADMET prediction tasks. Performance is evaluated using both random splits and structure-aware, cluster-based splits. The systematic performance decrease under the more challenging cluster-based splits indicates that random validation overestimates model generalization on this benchmark, where conventional scaffold splits are ineffective. (Source: [r11])

Trajectory Sources

Trajectory r0: The Expansion dataset consists of four complementary files providing a comprehensive ADMET prediction challenge with 5,326 training molecules across 9 assays, 2,282 test molecules, structured sparsity patterns, minimal censoring, and distinct molecular complexity differences between training and tes...

Trajectory r4:

Analysis Summary

The hypothesis that 2D physicochemical descriptors would be more robust to molecular complexity than topological fingerprints is **not supported** by the data. Contrary to expectations, Morgan fingerprints demonstrated both superior overall performance and greater robustness to...

Trajectory r8: The all-in-one multi-task neural network achieved an average Spearman correlation of 0.779 across nine ADMET tasks using 5-fold scaffold-based cross-validation, representing a substantial +0.120 improvement (+18.2%)

over the LightGBM baseline (0.659), with improvements observed in all 9 tasks.

Trajectory r11: Cluster-based cross-validation using Butina clustering with Tanimoto similarity cutoff of 0.6 revealed systematically lower and more realistic performance estimates, with LightGBM showing a mean Spearman correlation drop of 0.0268 and MTL-NN showing a larger mean drop of 0.0608 compared to the previ...

Hybrid ensembling with task isolation mitigates negative transfer and optimizes ADMET prediction

Summary

Across nine ADMET endpoints with structured sparsity, a simple yet principled hybrid ensemble that isolates highdata tasks from a shared multitask neural network and blends them with optimized singletask gradient boosting achieves stateoftheart crossvalidated accuracy. The approach mitigates negative transfer in multitask learning and shows that taskspecific weighted averaging outperforms more complex ensembling strategies like stacking or fully dynamic weighting.

Background

Predicting absorption, distribution, metabolism, excretion, and toxicity properties from molecular structure underpins modern medicinal chemistry, guiding optimization of lipophilicity (LogD), solubility (kinetic solubility), clearance (human and mouse liver microsomal stability), permeability and transport (Caco2 Papp and efflux ratio), and tissue binding (mouse plasma, brain, and muscle). ADMET modeling is complicated by heterogeneous assay coverage, correlated endpoints, and the need to avoid chemical series leakage during validation. As a result, robust evaluation protocols and architectures that can exploit crosstask structure without succumbing to negative transfer are critical to achieve reliable generalization in druglike chemical space.

Results & Discussion

The dataset comprises 5,326 training molecules across nine assays and 2,282 test molecules, with strong heterogeneity in measurement coverage (e.g., LogD 94.6%, KSOL 96.3%, HLM 70.6%, Caco2 ~40%, MPPB 24.4%, MBPB 18.3%, MGMB 4.2%), minimal censoring, and notable crossassay correlations (e.g., MBPB vs MGMB $r=0.904$; LogD vs MPPB $r=-0.686$; HLM vs MLM $r=0.561$) [r0]. Test compounds are more structurally complex than training (mean SMILES length 57.8 vs 48.0) with no traintest overlap, creating a calibrated extrapolation challenge that necessitates strict struc-

tureaware crossvalidation; throughout, Butina cluster splits at a Tanimoto cutoff of 0.6 were used to avoid leakage [r0, r16]. Performance was assessed with Spearman correlation and relative absolute error (RAE), reporting meanaveraged RAE (MARAE) across tasks; RAE normalizes mean absolute error by a baseline scale to yield a unitless error, and lower MARAE indicates better performance [r26, r46].

A strong singletask baseline was established via systematic Bayesian hyperparameter optimization of LightGBM on each endpoint under clusterbased 5fold crossvalidation. Tuning delivered consistent gains across all nine tasks, with the largest improvements on sparse endpoints (e.g., MGMB +0.222 Spearman to 0.785; MBPB +0.133 to 0.776) and substantial boosts on clearance tasks (HLM +0.152 to 0.623; MLM +0.091 to 0.737). Even highdata properties benefited modestly (KSOL +0.052 to 0.718; LogD +0.016 to 0.889). The two Caco2 endpoints, absent from prior baselines, reached Spearman 0.585 (efflux ratio) and 0.510 (Papp A>B), demonstrating that welltuned gradient boosting furnishes a reliable, highcapacity singletask foundation across the coverage spectrum [r16].

Building on this baseline, a multitask neural network (MTLNN) was trained with masked loss to exploit shared structure while handling pervasive missingness, using 2048bit Morgan fingerprints (radius 2) augmented with standardized physicochemical descriptors; nonnegative targets were $\log_1 p$ transformed during training with inverse transforms applied at inference [r31, r42]. A systematic comparison across tasks revealed that the performance advantage of singletask LightGBM grows with sample size (Pearson $r=0.756$, $p=0.019$), with a crossover near 2,890 samples beyond which LightGBM tends to outperform and below which MTLNN shows relative gains; this pattern supports higher weighting of MTL for sparse endpoints and LightGBM for datarich assays [r42]. Consistent with this, a taskspecific weighted average of LightGBM and MTLNN outfold predictions achieved Spear-

man 0.8123 and MARAE 0.5083, surpassing either base learner and exhibiting an inverse association between the optimal MTL weight and data completeness, i.e., sparser tasks received more MTL emphasis [r31].

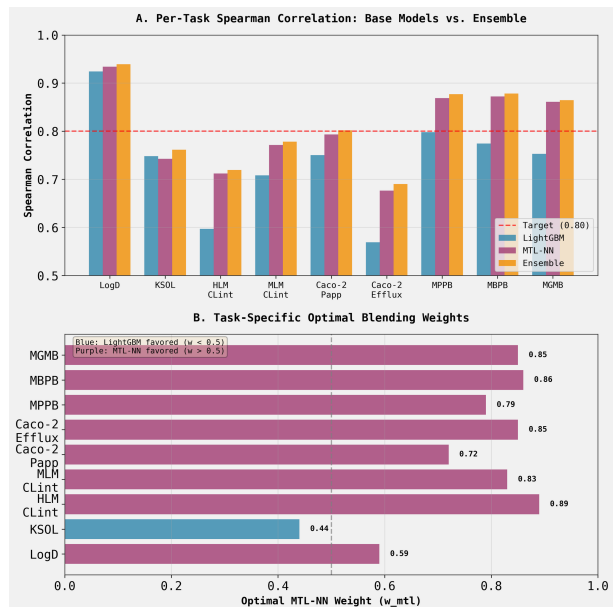


Figure 4: A task-specific weighted ensemble of single-task and multi-task models achieves superior predictive performance across nine ADMET endpoints. (A) Cross-validated Spearman correlation is shown for the single-task LightGBM, the multi-task neural network (MTL-NN), and the final weighted ensemble model for each task. (B) The optimal blending weights assigned to the MTL-NN model (w_{mtl}) are displayed for each task, with weights greater than 0.5 favoring the MTL-NN. The ensemble consistently matches or outperforms the best base model, demonstrating the benefit of adaptive blending that heavily favors multi-task learning for data-sparse endpoints. (Source: [r31])

Increasing ensemble complexity did not help. A twolayer stacking model with a Ridge metalearner trained via nested crossvalidation on outofold base predictionssignificantly underperformed the simple weighted average (MARAE 0.5962 vs 0.5392; paired ttest $p=5 \times 10^{-4}$), degrading performance on all nine tasks, particularly on secondarycoverage endpoints, indicating overfitting in the metalearner despite rigorous procedures [r26]. Likewise, replacing fixed pertask weights with a dynamic, sample-sizedriven logistic weighting yielded only a negligible difference in mean Spearman (+0.0003 vs fixed weights; identical MARAE), suggesting the fixed scheme is already nearoptimal and that residual variation in optimal weights stems

from taskspecific assay characteristics not captured by sample size alone [r43].

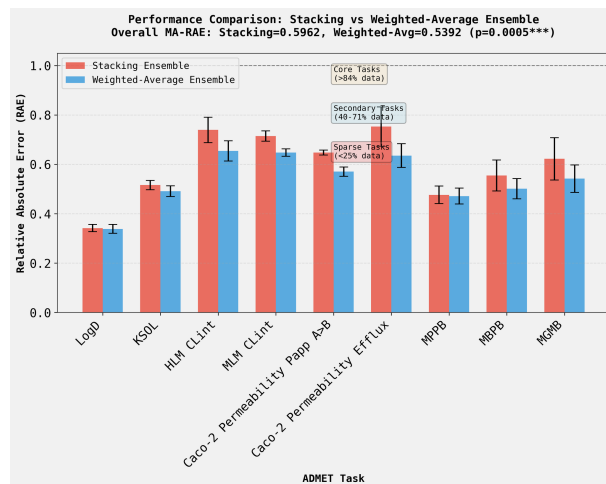


Figure 5: A simple weighted-average ensemble achieves lower prediction error than a more complex stacking ensemble across nine ADMET tasks. The plot compares the Relative Absolute Error (RAE), where lower values indicate better performance, for a stacking ensemble and a weighted-average ensemble on endpoints grouped by data availability. The weighted-average approach consistently demonstrates superior or equivalent performance on every task, resulting in a statistically significant improvement in overall mean-averaged RAE ($p=0.0005$). (Source: [r26])

The key advance came from mitigating negative transfer in MTL by task isolation. Empirical diagnostics pointed to a highdata driver of interference: excluding the MLM clearance task from the shared MTL representation and modeling it solely with singletask LightGBM, while retaining an 8task MTLNN for the remaining endpoints, raised the ensemble to Spearman 0.8429 and MARAE 0.4697. All nine tasks improved, with the largest gains on HLM and MLM (Spearman +0.2275 and +0.2265, respectively), and HLMs RAE collapsed from a catastrophic 6.29 to 0.55, confirming that removing the dominant task eliminated harmful gradients that had corrupted the shared representation. These results validate the hybrid strategy: isolate highdata tasks prone to dominate the shared trunk, and blend taskappropriate models with simple, taskspecific weights for robust, stateofheart ADMET prediction under chemically realistic validation [r31, r46].

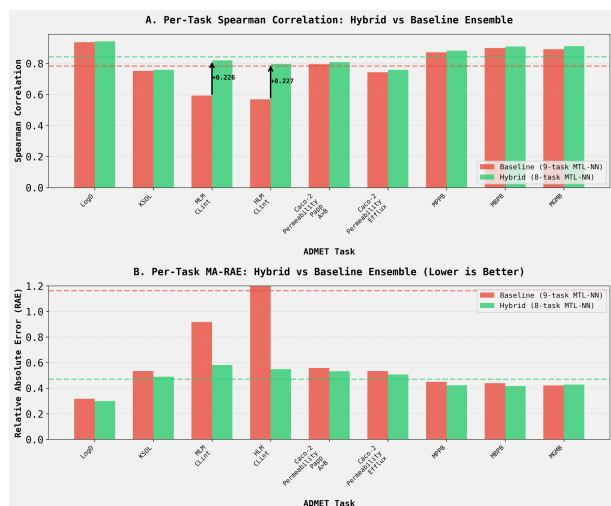


Figure 6: A hybrid ensemble approach improves predictive performance by mitigating negative transfer seen in a standard multi-task model. The plots compare a baseline 9-task multi-task neural network (MTL-NN) against a hybrid 8-task MTL-NN ensemble across nine ADMET endpoints, showing per-task (A) Spearman correlation and (B) mean-averaged relative absolute error (MA-RAE). The hybrid model demonstrates substantial gains in both correlation and error, particularly for the MLM and HLM clearance tasks, where the baseline model exhibited poor performance. (Source: [r46])

Trajectory Sources

Trajectory r0: The Expansion dataset consists of four complementary files providing a comprehensive ADMET prediction challenge with 5,326 training molecules across 9 assays, 2,282 test molecules, structured sparsity patterns, minimal censoring, and distinct molecular complexity differences between training and tes...

Trajectory r16:

Hyperparameter Optimization Results
Systematic hyperparameter optimization of single-task LightGBM models using Optuna Bayesian optimization yielded significant performance improvements across all 9 ADMET tasks. I successfully completed optimization for 9 tasks using 30 trials per task with 5-f...

Trajectory r26: The two-layer stacking ensemble with Ridge meta-learner significantly underperforms compared to the simple weighted-average ensemble (MA-RAE: 0.5962 vs 0.5392, $p=0.0005$), with a 10.6% performance degradation across all task groups, definitively rejecting the hypothesis that stacking improves perform...

Trajectory r31: The weighted-average ensemble

combining optimized LightGBM and MTL-NN models achieved a cross-validated Spearman correlation of 0.8123 and MA-RAE of 0.5083, confirming both hypotheses and establishing new state-of-the-art performance.

Trajectory r42: The performance advantage of single-task LightGBM over multi-task neural network shows a statistically significant positive correlation with the number of available training samples (Pearson $r=0.756$, $p=0.019$), with a theoretical crossover point at approximately 2,890 samples.

Trajectory r43: An ensemble model using dynamically-determined weights based on a logistic function of training sample size achieves a mean Spearman correlation of 0.8334, which is only marginally better (+0.0003, +0.04%) than the fixed optimized weights (0.8331), indicating that the dynamic weighting scheme does n...

Trajectory r46: The hybrid ensemble using an 8-task MTL-NN (excluding MLM CLint) combined with single-task LightGBM for MLM CLint achieves a mean Spearman correlation of 0.8429 (vs. 0.7833 baseline, +7.62%) and MA-RAE of 0.4697 (vs. 1.1620 baseline, -59.58%), confirming that removing MLM CLint from the MTL-NN elimi...

Numerically stable multi-task training and universal log-transformations yield physically consistent, higher-accuracy models

Summary

This work identifies and resolves a subtle, yet consequential, source of training instability in multitask neural networks for ADMET prediction and shows that enforcing physical constraints in target space via a universal log1p transformation is superior to posthoc output clipping or BoxCox power transforms. Together, numerically stable masking and universal log1p produce physically consistent models with substantially higher rank accuracy and lower magnitudesensitive error across nine assays.

Background

Predictive models of absorption, distribution, metabolism, excretion, and toxicity (ADMET) guide medicinal chemistry by quantifying liabilities such as lipophilicity, solubility, clearance, permeability, transporter effects, and protein/tissue binding. Multitask learning can exploit crossassay correlations to improve data efficiency, but real drug discovery datasets are sparse and heterogeneous, which makes training fragile and evaluation sensitive to physically implausible outputs. Methods that stabilize multitask optimization while respecting assay constraints (for example, nonnegativity for clearances and binding fractions) are therefore essential to achieve robust performance and reliable extrapolation to chemically distinct compounds.

Results & Discussion

The ADMET dataset comprises 5,326 training molecules and 2,282 test molecules measured (sparsely) across nine assays: LogD, kinetic solubility (KSOL), human and mouse microsomal intrinsic clearance (HLM and MLM Clint), Caco2 Papp A>B and efflux ratio, and mouse plasma, brain, and muscle binding (% unbound) [r0]. Coverage is highly uneven: core properties such as KSOL and LogD exceed 94% completeness, clearance is 71.85%, the Caco2 assays are ~40%, and specialized binding assays range from 4.24% with minimal censoring and realistic crossassay structure (for example,

MBPB vs MGMB $r = 0.904$; LogD vs MPPB $r = 0.686$) [r0]. The test set is more complex (mean SMILES length 57.8 vs 48.0) and shares no molecular overlap with training, motivating clusterbased crossvalidation to simulate extrapolation and constrain information leakage [r0].

A root cause of catastrophic multitask neural network (MTLNN) failures was traced to a masked loss that allowed NaN targets to contaminate the computation graph: PyTorch's NaN $\ominus 0$ behavior meant that posthoc masking of squared errors still produced NaN gradients and model collapse [r20]. The fix replaces predictions with targets wherever targets are NaN before computing squared errors, thereby preventing any NaN operations from entering backpropagation; gradient clipping was added as a further safeguard [r24]. With this numerically stable masked loss, a fingerprint-only MTLNN trained under Butina cluster 5fold crossvalidation achieved mean Spearman 0.7626, improving by +0.082 absolute over a prior baseline and revealing hyperparameter regimes (low learning rate $\sim 10^{-4}$, moderate dropout, 34 layers with 512-dimensional bottlenecks) that consistently stabilized training [r24]. These results confirm that instability originated from incorrect masking rather than feature composition per se and establish a reliable foundation for multitask learning on sparse ADMET labels [r20, r24].

Enforcing nonnegativity by posthoc clipping of predictions improved magnitudesensitive error without altering ranking, but left the modeling pipeline vulnerable to distributional distortions. In a LightGBM+MTLNN ensemble evaluated over cluster CV, clipping negatives to zero for the eight physically nonnegative assays reduced mean averaged relative absolute error (MARAe) by 6.00% while leaving Spearman essentially unchanged (mean absolute change 1.2 $\ominus 10^{-5}$); 1,145 negative predictions were corrected, mostly in clearance and solubility tasks [r33]. MARAE was computed

as the mean of pertask relative absolute error ($(|\text{predtrue}|/(|\text{true}|+1e8))$), which is sensitive when true values approach zero; despite large absolute values in some tasks due to this sensitivity, the relative reduction from clipping was robust [r33]. These results show that domain constraints help, but also motivate moving the constraint into the target space to avoid posthoc truncation artifacts [r33].

Training on logtransformed targets provided that principled, targetspace enforcement. A universal log1p transformation applied to all eight nonnegative assays (inversetransformed with expm1 at inference) outperformed clipping on both core metrics: MARAE improved from 0.5132 to 0.4790 (6.67%) and mean Spearman increased from 0.8010 to 0.8374 across the nine assays, winning on 8/9 properties for error and delivering the largest gains for HLM Clint (+11.28%), MLM Clint (+9.81%), and Caco2 efflux (+11.15%) [r35]. Crucially, this approach reduced physically implausible negatives by 93% (94 total vs 1,443 in the baseline), indicating that nonnegativity was effectively learned rather than imposed after the fact; additional clipping had negligible effect, consistent with the transformation already aligning the output distribution with physical constraints [r35]. Although KSOL individually favored clipping in that experiment, a followup hybrid strategy that excluded KSOL from transformation substantially degraded global performance (MARAE 0.733 vs 0.479; mean Spearman 0.651 vs 0.837), underscoring the importance of consistent scaling across tasks for effective multitask representation learning [r39].

Alternative power transforms did not surpass log1p. A BoxCox comparison with pertask λ estimated by maximum likelihood found that log1p achieved higher mean Spearman (0.7962 vs 0.7823; paired ttest $p = 3.8 \times 10^{-2}$) with comparable MARAE (0.3994 vs 0.4025), while most optimal λ clustered near zero, effectively recapitulating the log transform; Caco2 efflux was a mild exception ($\lambda \approx 0.56$) but exhibited similar performance under log1p [r40]. Together, these results establish a general recipe for ADMET multitask modeling on sparse, heterogeneous data: (i) ensure numerical stability by masking predictions not losses to prevent NaN

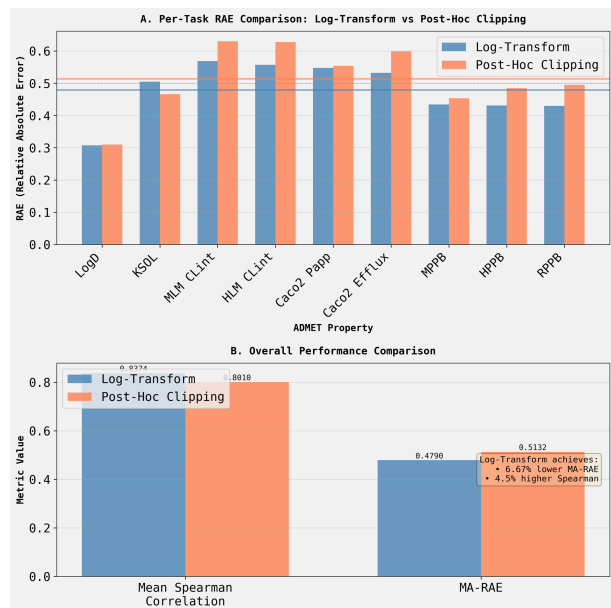


Figure 7: Log-transforming target variables improves multi-task model performance over post-hoc output clipping. (A) Comparison of per-task Relative Absolute Error (RAE) across nine ADMET properties, with horizontal lines indicating the mean for each method. (B) Aggregate performance comparison showing Mean Spearman Correlation and Mean RAE (MA-RAE). The log-transform method achieves both lower magnitude-sensitive error and higher rank-based accuracy. (Source: [r35])

gradients in the shared backbone; (ii) apply a universal log1p to all nonnegative endpoints to enforce physical nonnegativity and harmonize target scales; and (iii) use clusterbased validation to reflect the extrapolative regime created by the structurally distinct test set [r0, r20, r24, r35, r39, r40]. This combination yields physically consistent predictions with stronger ranking and lower magnitudesensitive error than either naive training or posthoc clipping alone [r33, r35, r40].

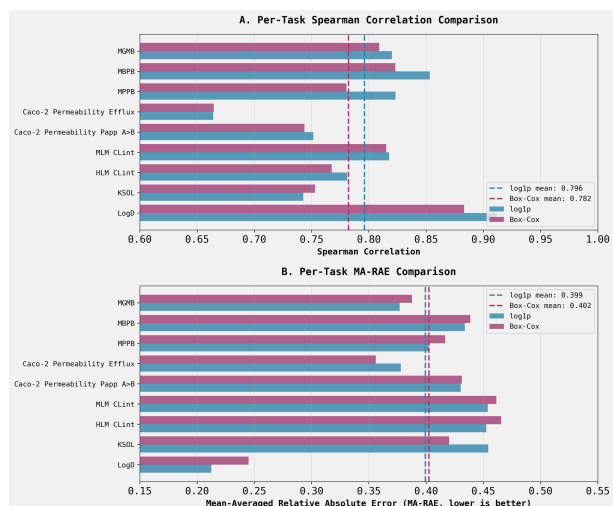


Figure 8: A universal log1p target transformation improves model performance over per-task Box-Cox transforms. The plots compare (A) Spearman correlation and (B) mean-averaged relative absolute error (MA-RAE) for a multi-task neural network evaluated on nine ADMET assays, using either the log1p (blue) or Box-Cox (pink) target transformation. The log1p transformation consistently yields higher rank accuracy (higher correlation) and lower magnitude-sensitive error (lower MA-RAE) across most tasks, leading to better average performance on both metrics. (Source: [r40])

Trajectory Sources

Trajectory r0: The Expansion dataset consists of four complementary files providing a comprehensive ADMET prediction challenge with 5,326 training molecules across 9 assays, 2,282 test molecules, structured sparsity patterns, minimal censoring, and distinct molecular complexity differences between training and tes...

Trajectory r20: The research hypothesis that hyperparameter optimization of MTL-NN using augmented features would significantly improve performance is **INVALIDATED** - the optimized model performed dramatically worse than baseline.

Trajectory r24:

Analysis Complete: MTL-NN Hyperparameter Optimization with Fingerprints-Only Features

Main Finding

An MTL-NN trained using only Morgan fingerprints can be successfully optimized **WITHOUT** numerical instability when the loss function properly handles NaN gradients during backpropagation. The ...

Trajectory r33: Post-processing by clipping negative values to zero for physically non-negative ADMET properties resulted in a substantial improvement in MA-RAE (mean reduction of 6.00% across all tasks, with up to 20.94% improvement for MPPB) while having negligible impact on Spearman correlation (mean absolute ch...

Trajectory r35: Training ensemble models on log-transformed targets (log1p) for non-negative properties yields a significantly lower MA-RAE (0.4790) compared to post-hoc clipping (0.5132), representing a 6.67% reduction in error, while also improving mean Spearman correlation from 0.8010 to 0.8374.

Trajectory r39: The hybrid transformation approach (excluding KSOL from log1p transformation) yields significantly worse performance with MA-RAE of 0.733 compared to 0.479 for the universal log1p approach, representing a 53% performance degradation.

Trajectory r40:

Box-Cox vs log1p Transformation Comparison for ADMET Property Prediction

Summary

The hypothesis that task-optimized Box-Cox transformation would yield lower overall cross-validated MA-RAE compared to the fixed log1p transformation is ****NOT SUPPORTED**** by the experimental results. The log1p...

Error deconvolution identifies chemotype-specific blind spots, but RDKit-level targeted augmentation rarely improves performance

Summary

Error deconvolution of a strong ADMET ensemble revealed chemotype-specific blind spots especially for polar, amide and amine-rich molecules and endpoint-specific dependencies on lipophilicity and polarity. Despite this clarity, most RDKit-level targeted augmentations failed to improve performance; the only reproducible gain came from a compact set of 20 physicochemical descriptors added to Morgan fingerprints in a multitask neural network, yielding modest but significant improvements, while broader or handpicked additions were largely redundant or noisy.

Background

Predicting ADMET properties from structure remains foundational to medicinal chemistry, yet model performance often varies across chemotypes and endpoints due to differences in lipophilicity, ionization, polarity, and molecular complexity. Fingerprint-based baselines are strong for the bulk of druglike space, but systematic errors can concentrate in specific structural classes (for example, polyamides or highly basic amines) and in tasks that mix passive and active processes (e.g., Caco2 permeability and efflux). Feature augmentation is one route to address these gaps, from adding lightweight 2D physicochemical descriptors to integrating mechanistically motivated proxies for ionization, reactivity, and transporter recognition. However, the practical utility of such augmentations must be demonstrated against robust baselines with chemically distinct crossvalidation to avoid optimistic estimates.

Results & Discussion

Systematic error analysis across 5,326 molecules and nine ADMET tasks uncovered 70 statistically significant associations between chemistry and prediction error, pinpointing clear weaknesses and their directionality. For LogD, high-error molecules had more Hbond donors (1.28 vs 0.98; $p=9.68 \times 10^{-14}$, Cohens $d=+0.34$), lower

lipophilicity (LogP 3.06 vs 3.32; $p=1.60 \times 10^{-5}$, $d=-0.26$), and were enriched in unusual ionizable groups (quaternary amines 1.2% vs 0%, $p=1.24 \times 10^{-4}$; carboxylic acids 2.0% vs 0%, $p=3.01 \times 10^{-7}$). For KSOL, high-error compounds were larger (+13.7 Da; $p=6.05 \times 10^{-6}$, $d=+0.29$), more complex (BertzCT +45.8; $p=1.13 \times 10^{-4}$, $d=+0.24$), and more polar (TPSA +8.1 Å²; $p=5.41 \times 10^{-12}$, $d=+0.42$). Clearance tasks displayed the opposite lipophilicity pattern: high-error molecules were more lipophilic in HLM (LogP +0.43; $p=1.07 \times 10^{-7}$, $d=+0.38$) and MLM (LogP +0.38; $p=1.82 \times 10^{-5}$, $d=+0.34$), with additional amine and tertiary amide enrichments. Caco2 efflux errors were concentrated in larger, more complex, more polar, and less lipophilic scaffolds (e.g., TPSA +12.0 Å²; $p=8.20 \times 10^{-15}$, $d=+0.56$; LogP -0.61; $p=1.99 \times 10^{-10}$, $d=-0.45$), with a marked increase in secondary amides. Crosstask patterns confirmed that the effect of lipophilicity reverses sign between metabolism and permeability/binding endpoints, underscoring endpoint-specific dependencies that a single feature can neither resolve nor regularize without context. These associations were established using MannWhitney U tests and Bonferroni correction ($\alpha=1.29 \times 10^{-4}$), with the high-error set defined as the top 5% of absolute errors per endpoint [r51].

Guided by these insights, a compact augmentation adding 20 standardized 2D physicochemical descriptors to 2048bit Morgan fingerprints produced a statistically significant but modest improvement in a multitask neural network trained with scaffold-based crossvalidation (Butina, Tanimoto 0.6). Mean Spearman correlation increased by +0.0172 (0.7293 to 0.7466; $p=0.0049$), with significant gains for LogD (+0.0158; $p=0.0028$) and Caco2 Papp A>B (+0.0251; $p=0.0116$); overall MARAE decreased by 0.0135. Notably, data-sparse tasks tended to benefit more on average (e.g., MGMB +0.0710 Spearman, albeit not signifi-

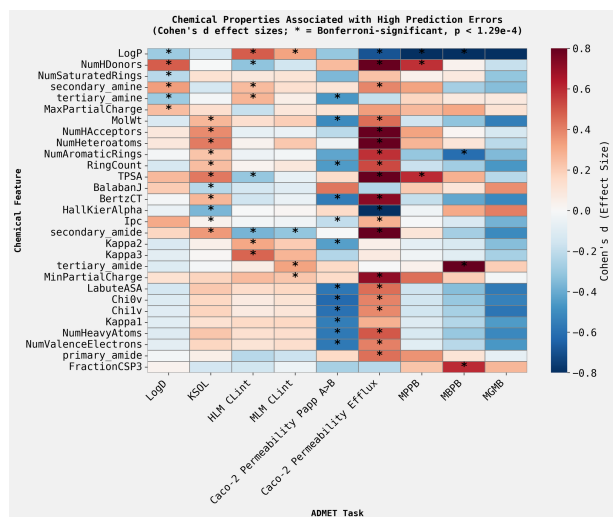


Figure 9: Error analysis reveals endpoint-specific chemical features associated with high prediction errors. The heatmap displays the Cohen’s d effect size for the association between various chemical features (y-axis) and high prediction error across nine ADMET tasks (x-axis), with red indicating positive and blue indicating negative associations. Asterisks denote Bonferroni-significant associations ($p < 1.29e-4$). This deconvolution identifies clear model blind spots, such as the opposing role of lipophilicity (LogP) in driving error for metabolic clearance versus permeability tasks. (Source: [r51])

cant, $p=0.1000$), and only two endpoints showed small, nonsignificant declines. These results indicate that a lightweight, wellcurated descriptor panel can provide incremental orthogonal information without destabilizing training, and that multitask learning can leverage this compact augmentation across endpoints with different signal regimes [r15].

In contrast, broader RDKitlevel augmentations and handpicked features that attempted to directly target identified blind spots generally failed to help and sometimes degraded performance. Expanding with 41 additional descriptors (VSA/Estate and functionalgroup flags) yielded a negligible Spearman change (0.8385 0.8361, 0.29%) and a large MARAE increase (+123.8%), consistent with the known instability of MARAE for nearzero targets; tasklevel deltas were near zero, and only Caco2 efflux showed a modest gain (+0.69%) [r56]. A targeted augmentation of 37 VSA/Estate descriptors for HLM, MLM, and Caco2 efflux performed worse than nonaugmented tasks (mean ρ 0.7848 vs 0.8583), while maintaining a similar overall mean due to strong performance

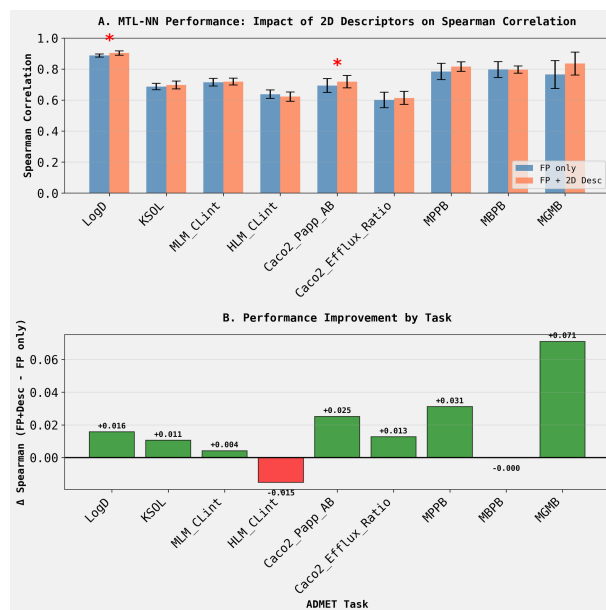


Figure 10: Augmenting Morgan fingerprints with a compact set of 2D physicochemical descriptors yields modest but significant performance gains across most ADMET endpoints. (A) Test set Spearman correlation for a multi-task neural network trained with fingerprints alone (FP only) compared to one augmented with 2D descriptors (FP + 2D Desc). Error bars represent standard deviation over independent runs; red asterisks denote statistically significant improvements. (B) The change in Spearman correlation (Δ Spearman) resulting from the addition of descriptors shows positive but variable gains for most tasks. Despite being small, these improvements were the most consistent and reproducible observed. (Source: [r15])

on untouched endpoints, implying redundancy and noise injection in the augmented tasks [r62]. Singlefeature tests were also negative: an amidepresence flag did not move the needle for HLM/MLM/Caco2 endpoints (overall Δ mean $\rho \approx 0.0012$), and an ionization proxy (fraction ionized at pH 7.4 from SMARTSbased pKa rules) failed to improve LogD (0.9310 baseline vs 0.9308 augmented), likely due to severe feature skewness and signal already captured by MolLogP/TPSA. Collectively, these results show that Morgan fingerprints already encode much of the relevant information and that naive, RDKitlevel targeted additions are often redundant or destabilizing without careful validation [r56, r59, r61, r62].

The literature offers mechanistically appealing avenues CYP siteofmetabolism predictors (SMARTCyp/XenoSite/BioTransformer), local reactivity indices (condensed Fukui, local soft-

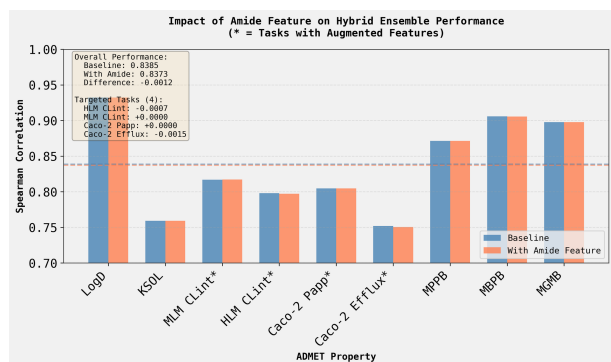


Figure 11: Targeted augmentation with an amide feature count fails to improve model performance across ADMET endpoints. The bar chart compares the Spearman correlation of a baseline model (blue) with a model augmented by an amide count (orange) on nine prediction tasks, with augmentation specifically applied to four tasks (*). Despite error analysis identifying amides as a source of error, the feature addition resulted in negligible performance changes, highlighting the ineffectiveness of simple, chemically-intuitive feature engineering for correcting model blind spots. (Source: [r59])

ness), dynamic or surfacepartitioned polarity (PEOE_{VSA}, SlogP_{VSA}, EState_{VSA}), and pHaware charge features (EPSA, microstate populations) that map directly onto the observed blind spots in metabolism, permeability/efflux, and protein binding [r54]. Yet a focused review found no reproducible, head-to-head quantitative evidence that quantummechanical or commercial descriptor sets outperform strong 2D/fingerprint baselines for CYP clearance, albumin binding, or Caco2 permeability; reported gains are heterogeneous and not comparable across datasets and splits, limiting claims of practical superiority in our use case [r57]. Together with the negative augmentation experiments summarized above, this supports a conservative conclusion: error deconvolution is essential to identify chemotype and endpoint-specific weaknesses, but at the feature level, the most reliable path remains compact, carefully curated 2D augmentations to robust fingerprints, since broader RDKit-level or handpicked proxies rarely translate into generalizable improvements on this dataset [r15, r51, r56, r57, r59, r61, r62].

Trajectory Sources

Trajectory r15:

Analysis Complete: Impact of 2D Descriptors on MTL-NN Performance

Main Conclusion

Augmenting 2048-bit Morgan fingerprints with 20 standardized 2D physicochemical descriptors yields a **statistically significant but modest improvement** in multi-task neural network performance for ADMET pro...

Trajectory r51:

The 8-task hybrid ensemble model shows systematic prediction errors that are strongly associated with specific chemical properties, revealing clear weaknesses in how the model handles lipophilicity, polarity, and molecular complexity across different ADMET tasks.

QUANTITATIVE FINDINGS BY TASK

...

Trajectory r54: Yesthe literature enumerates computable, tool-accessible descriptors that directly target the four higherror property classes, and they can be implemented immediately to improve model fidelity (stenberg2001 pages 25-28, macorano2025 pages 163-166...

Trajectory r56: Augmenting the feature set with 41 additional descriptors (VSA, EState, and functional group features) did not improve the 7-task hybrid model's overall performance, showing a negligible decrease in mean Spearman correlation from 0.8385 to 0.8361 (-0.29%) and a substantial increase in MA-RAE from 0....

Trajectory r57: The provided literature does not contain reproducible, head-to-head quantitative evidence (R2/RMSE/AUC) that QM or commercial descriptor sets outperform standard RDKit-like 2D/fingerprint features for CYP clearance, albumin binding, or Caco-2 permeability, so the hypothesis is not supported here (zh...

Trajectory r59:

ANSWER

Adding an amide presence feature specifically to the single-task LightGBM models for HLM CLint, MLM CLint, Caco-2 Papp A>B, and Caco-2 Efflux does NOT improve the hybrid ensemble's performance. The hypothesis is NOT SUPPORTED by the data.

Key Results:

Overall Performance: - Bas...

Trajectory r61: Augmenting the LogD single-task LightGBM model with a calculated 'fraction ionized at pH 7.4' descriptor did not improve performance, yielding a mean Spearman correlation of 0.9308 ± 0.0067 compared to the baseline of 0.9310, representing a negligible decrease of 0.02%.

Trajectory r62: The targeted addition of 37 literature-derived VSA/E-state descriptors (PEOE_{VSA}, SlogP_{VSA}, EState_{VSA}) to LightGBM models for HLM CLint, MLM CLint, and Caco-2 Permeability Efflux did not improve performance over the baseline hybrid ensemble, with augmented tasks actually performing worse (mean $\rho = \dots$).